

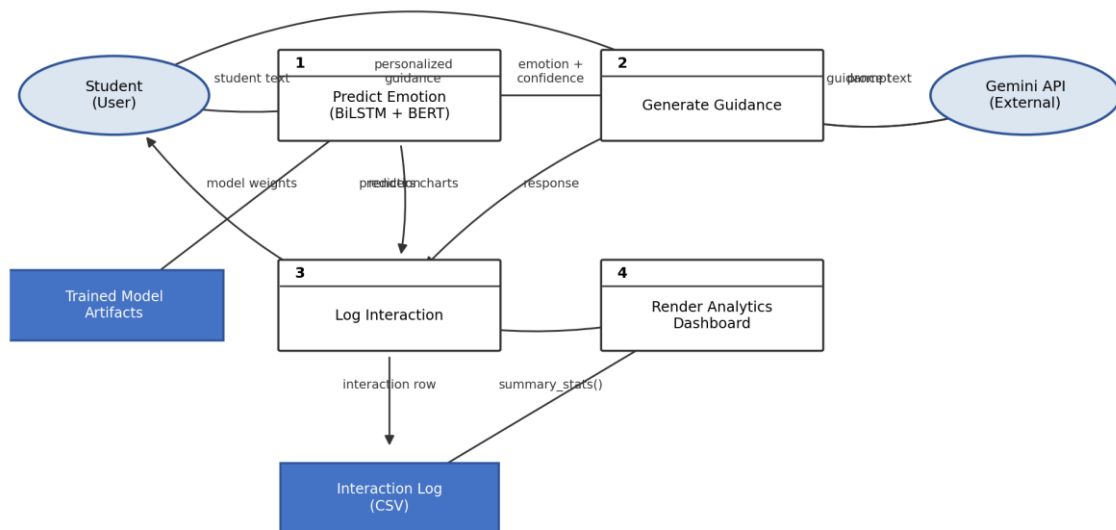
Data Flow Diagram

Date	12 July 2026
Team ID	
Project Name	Emotion Detection & Learning Support Engine
Maximum Marks	2 Marks

Data Flow Diagram

AI Learning Assistant — Data Flow Diagram

Data Flow Diagram — AI Learning Assistant



Description: The Student (external entity) submits free-text input, which Process 1 (Predict Emotion) classifies using BiLSTM and BERT models loaded from the Trained Model Artefacts data store. The predicted emotion and confidence flow to Process 2 (Generate Guidance), which exchanges a prompt and response with the Gemini API (external entity) and returns personalised guidance to the student. Both the prediction and the response are passed to Process 3 (Log Interaction), which writes a row to the Interaction Log data store. Process 4 (Render Analytics Dashboard) reads summary statistics back from that data store and renders charts back to the student.